Computer Networks LAB-01

Experiment Title :→

Study of LAN Topology, Network Devices and Address Configuration.

Problem Statement

Objective :→ To study different LAN topologies, identify network devices, analyze connection types, measure network speed, and obtain system network details.

Requirements:
- Computer/Laptop
- Internet Connection
- LAN Cable / Wi-fi
- Command Prompt

A Local Areal Network (LAN) connects devices within a limited area such as a college or office. Network topology defines how devices are arranged and connected.

Common topologies :
- Star Topology.
- Bus Topology.
- Mesh Topology.

Procedure :

(1) Draw the Network Topology:

Hybrid Hierarchical Topology which combines:

① Tree Topology
② Extended star topology
③ Ring topology (server room)

The network hierarchical layer :→

1. Core layer
2. Distribution layer
3. Access layer.

(i) Server Room - Mesh Topology :-

All switches are interconnected using fiber uplink. Core switch - 1 and core switch-2 directly connected to each other and to both Distribution switch-1 and Distribution switch-2. Router / Firewall server, and Network monitoring PC connected to core switches using Ethernet LAN. This forms the central core of the compus.

(ii) Academics 1st, 2nd, 3rd floor - Star topology :

Each floor connects to the server room via fiber uplink from core switches to the floor switches. whithin each floor, switch-1 and switch-2 Connect to Access points AP-1 to AP-6 using Ethernet LAN or wifi. Each AP has a dedicated Connection to the Central switch.

(iii) Girls Hostel - (Bus Topology):-

Hostel switch connects to server room via fiber uplink. APs tap into the common bus cable.

(iv) Boys Hostel :- (Ring topology):-

APs connected in a closed circular loop. Each AP links to two neighbors.

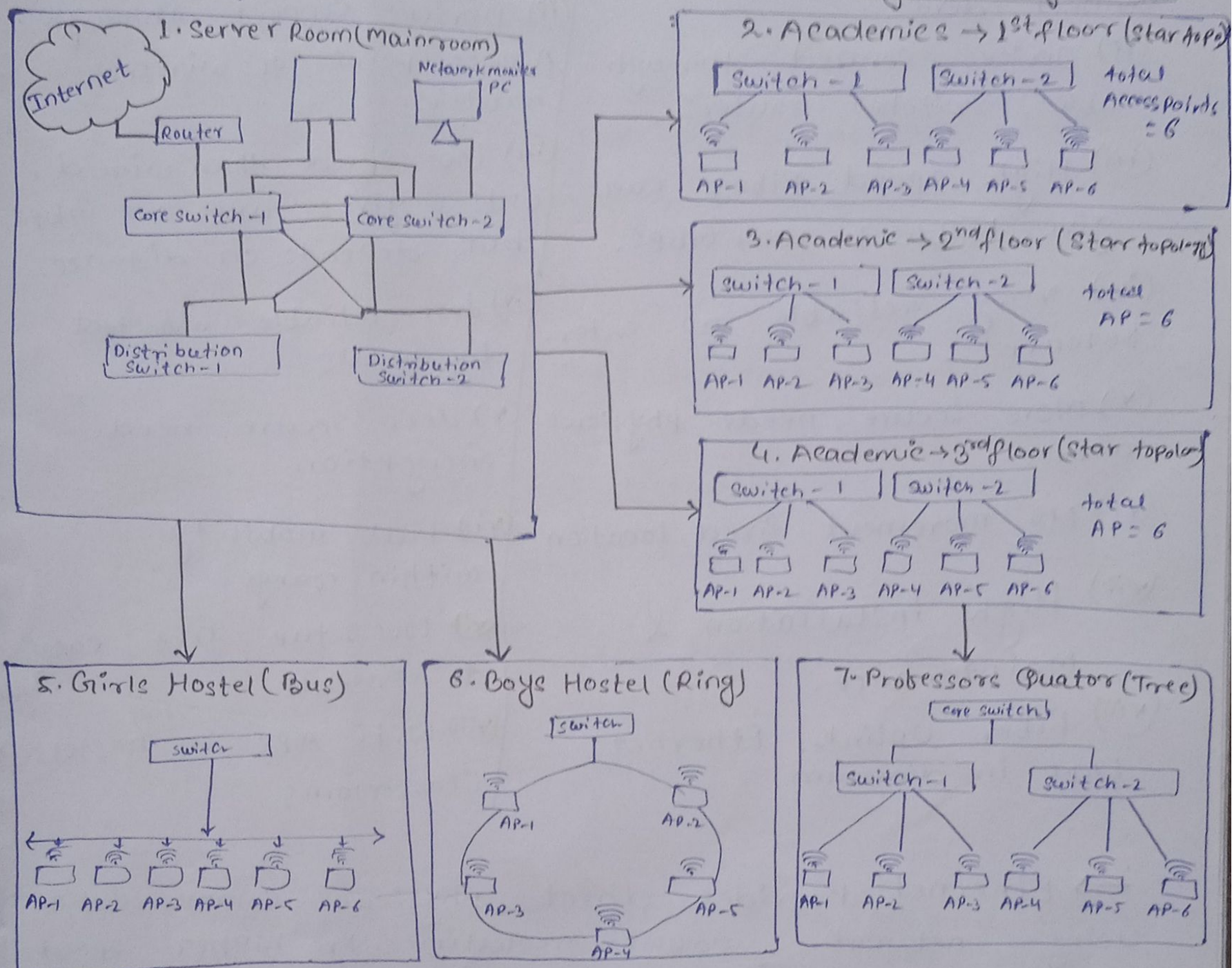
(v) Professor Quater - Tree Topology :-

Core Switch connects to server room via fiber Uplink. Sub-Switches connect to APs via Ethernet.

(vi) Overall Campus - (Hybrid Topology):

Mix of Mesh, Star, Bus, Ring, Tree in one Network.

College CAMPUS LAN - multiple topologies diagram



(8). OVerall Campus (HYbrid ToPology)

- Server Room (mesh)
- Academics 1st floor (star)
- Academics 2nd floor (star)
- Academic 3rd floor (star)
- Girls Hostel (Bus)
- Boys Hostel (Ring)
- Professors (Tree).

```
ows IP Configuration


rnet adapter Ethernet:

edia State . . . . . : Media disconnected
onnection-specific DNS Suffix  . :

ernet adapter Ethernet 2:

Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::8a97:b3d5:4864:820%9
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

reless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

reless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

ireless LAN adapter WiFi:

Connection-specific DNS Suffix  . :
IPv6 Address. . . . . : 2409:40e5:b6:8c27:c5ca:3ded:c601:4125
Temporary IPv6 Address. . . . . : 2409:40e5:b6:8c27:711f:f17e:3418:6b99
Link-local IPv6 Address . . . . . : fe80::f722:19be:1dbf:bbc2%11
IPv4 Address. . . . . : 10.149.233.229
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::6ca7:2aff:fe97:f345%11
                            10.149.233.106

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

Run new task

CPU
10% 4.17 GHz

Memory
4.3/5.8 GB (74%)

Disk 0 (C: D: E:)
SSD (NVMe)
6%

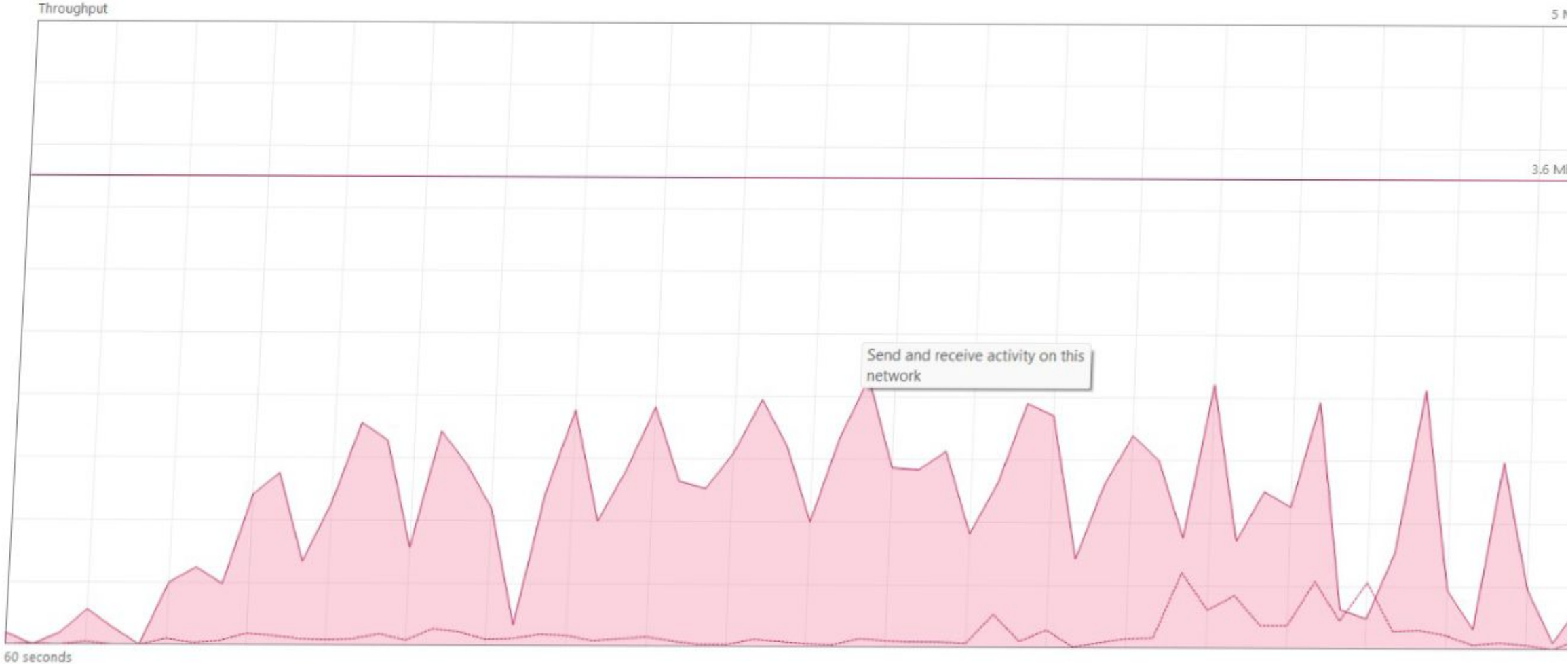
Ethernet
Ethernet 3
S: 0 R: 0 Kbps

Wi-Fi
Wi-Fi 2
S: 88.0 R: 368 Kbps

GPU 0
AMD Radeon(TM) 6...
2% (46 °C)

Wi-Fi

MediaTek Wi-Fi 6 MT7921 Wireless LAN C



Send
88.0 Kbps

Receive
368 Kbps

Adapter name:	Wi-Fi 2
SSID:	Redmi 9A
Connection type:	802.11n
IPv4 address:	192.168.43.17
IPv6 address:	2401:4900:1688:cda6:7aad:d636:2a48:c768
Signal strength:	

```
PS C:\Users\gauta\Desktop\DSA-C-> getmac

Physical Address    Transport Name
=================== ==========================================================
2C-98-11-76-BB-0B   \Device\Tcpip_{760FB142-AC12-484E-A8E2-6C816B6DA4FA}
C4-C6-E6-26-10-DA   Media disconnected
0A-00-27-00-00-06   \Device\Tcpip_{38EAA279-0FBD-481D-8972-70CD8A3321C1}
PS C:\Users\gauta\Desktop\DSA-C->
```

Windows PowerShell

```
dows PowerShell
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tall the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

C:\Users\USER_1> ipconfig

ndows IP Configuration


hernet adapter Ethernet:

 Media State . . . . . . . . . . . : Media disconnected
 Connection-specific DNS Suffix  . :

thernet adapter Ethernet 2:

 Connection-specific DNS Suffix  . :
 Link-local IPv6 Address . . . . . : fe80::8a97:b3d5:4864:820%9
 IPv4 Address. . . . . . . . . . . : 192.168.56.1
 Subnet Mask . . . . . . . . . . . : 255.255.255.0
 Default Gateway . . . . . . . . . :

thernet adapter Ethernet 3:

  Connection-specific DNS Suffix  . :
  IPv6 Address. . . . . . . . . . . : 2401:4900:b460:1e44:4949:b472:7b3c:fc79
  Temporary IPv6 Address. . . . . . : 2401:4900:b460:1e44:3c9e:82e7:a1ce:f2f6
  Link-local IPv6 Address . . . . . : fe80::c193:b6e4:35c1:5925%59
  IPv4 Address. . . . . . . . . . . : 10.15.230.218
  Subnet Mask . . . . . . . . . . . : 255.255.255.0
  Default Gateway . . . . . . . . . : fe80::78e2:37ff:fe5a:3abd%59
                                      10.15.230.124

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:
```

Difference b/w wired and wireless speed?

Ans→

Wired Network	Wireless Network
(i) Uses physical cables likes Ethernet, Fiber optical to connect devices.	(i) Uses radio waves, Wi-fi, Blue tooth. NO physical cables
(ii) Devices connect through cables, switches, routers..	(ii) Devices connect through Access points or wireless routers.
(iii) High speed fiber can give 1Gbps to 100+ Gbps.	(iii) low speed than wired. wifi 6 gives upto 9.6 Gbps but depends on distance.
(iv) very reliable, no interference	(iv) less reliable affected by walls.
(v) More secure, needs physical access	(v) less secure, need encryption.
(vi) NO movement fixed location	(vi) Full mobility within range.
(vii) High installation & maintenance.	(vii) Low setup cost, easy to install
(viii) Fiber Uplink, Ethernet LAN in diagram.	(viii) Wifi APs in hostels Classrooms.

→ Ethernet Prorides direct physical Communication using network cables, resulting in higher speed & stable Performance.

→ Wireless Connections uses radio waves which are affected by interference, distance, obstacles & multiple connected users.

Role of Subnet mask in network ?

Ans→ A Subnet mask seperates an IP address into network and host parts, helping identify devices within a network. it also enables subneetting, efficient muting, and reduces network traffic by dividing large network into smaller ones.

Observation :→

During the study of the college network infrastructure:

(1) A centralized server room was used as network backbone.

(2) Two L3 switches were Conneeted in redundant cyclic structure.

(3) Managed TP-Link PoET Switches were used in network cabinets.

Adrantage of Existing Network -

(i) Scalable architeture.
(ii) Centralized managent
(iii) Supports PoET devices.
(iv) Easy expansion of floore & classroom.
(v) Support both wireless & wired.
(vi) Redundant link for failouer.

Conclusion :→

The obserd campous LAN follow a Hybrid heerarchical topology consist of tree, extended star, Righg, the network uses managed PoE Switches, wireless access point, Ethernet-Port & centralized Server infrastructure to Pronide efficient wired & wireless Communication. across the campous.